CODING SHUTTLE

Kubernetes

Config Map and Secrets

ConfigMap

A ConfigMap is a Kubernetes object used to store non-confidential configuration data in key-value pairs. ConfigMaps are essential for managing configuration settings separately from the application code, making your applications more flexible and portable.

```
kubectl create configmap my-config --from-literal=db_host=database.example.com --from-literal=db_port=5432
```

Or using a file:

```
kubectl create configmap my-config --from-file=app.properties
```

Using Config Maps

```yaml
apiVersion: v1
kind: ConfigMap
metadata:
  name: my-config
data:
  app.properties: |
    db_host=database.example.com
    db_port=5432
  log_level: DEBUG
```

```yaml
apiVersion: v1
kind: Pod
metadata:
  name: my-pod
spec:
  containers:
  - name: my-container
    image: my-image
    env:
    - name: DB_HOST
      valueFrom:
        configMapKeyRef:
          name: my-config
          key: db_host
    - name: DB_PORT
      valueFrom:
        configMapKeyRef:
          name: my-config
          key: db_port
```

Secrets

Kubernetes Secrets are used to store sensitive information such as passwords, tokens, and certificates in a secure way.

Secrets can be used by Kubernetes deployments, pods, and other resources that require this sensitive data.

Secrets can be passed as:

- Environment variables in a container
- Sensitive Files in a container
- Command-line arguments in a container

Creating Secrets

```
kubectl create secret generic my-secret --from-literal=password=abc123
```

```
kubectl create secret generic my-secret --from-file=ssh-privatekey=/path/to/privatekey --from-file=ssh-publickey=/path/to/publickey
```

```yaml
apiVersion: v1
kind: Secret
metadata:
  name: my-secret
type: Opaque
data:
  DB_USERNAME: bXl1c2Vy   # Base6
  DB_PASSWORD: bXlwYXNzd29yZA==
```

Using Secrets

```yaml
apiVersion: v1
kind: Secret
metadata:
  name: my-secret
type: Opaque
data:
  DB_USERNAME: bXl1c2Vy   # Base6
  DB_PASSWORD: bXlwYXNzd29yZA==
```

```yaml
    app: my-app
spec:
  containers:
    - name: my-container
      image: my-image:latest
      env:
        - name: DB_USERNAME
          valueFrom:
            secretKeyRef:
              name: my-secret
              key: DB_USERNAME
        - name: DB_PASSWORD
          valueFrom:
            secretKeyRef:
              name: my-secret
              key: DB_PASSWORD
```